

MGT 595: Quantitative Investing

Problem Set 2

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1 Question 1: Mean–Variance Portfolios — 10 Industries

Data and conventions. The data are monthly value-weighted returns on ten U.S. industry portfolios from July 1926 to June 2026, giving $T = 1200$ observations on $N = 10$ industries, with no missing values. All figures are in percent per month unless stated otherwise: a mean of 1.0584 means 1.0584% per month. The risk-free rate is the sample mean of the monthly risk-free column, $r_f = 0.2701\%$ per month. Short sales are unrestricted, so writing μ for the vector of sample means, Σ for the sample covariance matrix and $\mathbf{1}$ for a vector of ones,

$$w_{\text{MVP}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}, \quad w_{\text{TAN}} = \frac{\Sigma^{-1}(\mu - r_f\mathbf{1})}{\mathbf{1}'\Sigma^{-1}(\mu - r_f\mathbf{1})}, \quad (1)$$

with $\mu_p = w'\mu$, $\sigma_p = \sqrt{w'\Sigma w}$ and $S_p = (\mu_p - r_f)/\sigma_p$.

1.1 Question 1(a): MVP, Tangency Portfolio and the Efficient Frontier

Table 1: Industry sample moments and the resulting portfolio weights from equation (1). Weights sum to one in each column. The last three rows report the mean, standard deviation and Sharpe ratio of each portfolio.

Industry	Monthly (%)		Weight	
	Mean	Std. Dev.	MVP	Tangency
NoDur	0.9402	4.5595	0.7188	0.6645
Durbl	1.1822	8.2136	−0.0142	0.1340
Manuf	1.0385	6.2008	−0.1905	−0.2242
Enrgy	1.0292	6.4095	0.1527	0.2808
HiTec	1.1699	7.1424	−0.0621	0.1865
Telcm	0.8124	4.6606	0.4889	0.1525
Shops	1.0342	5.7802	−0.0137	0.0725
Hlth	1.0680	5.5005	0.1174	0.3530
Utils	0.8874	5.4351	0.1293	0.0721
Other	0.9237	6.3555	−0.3267	−0.6916
<i>Portfolio mean</i> (%/mo)			0.8672	1.0584
<i>Portfolio std. dev.</i> (%/mo)			3.8346	4.4063
<i>Sharpe ratio</i> (monthly)			0.1557	0.1789

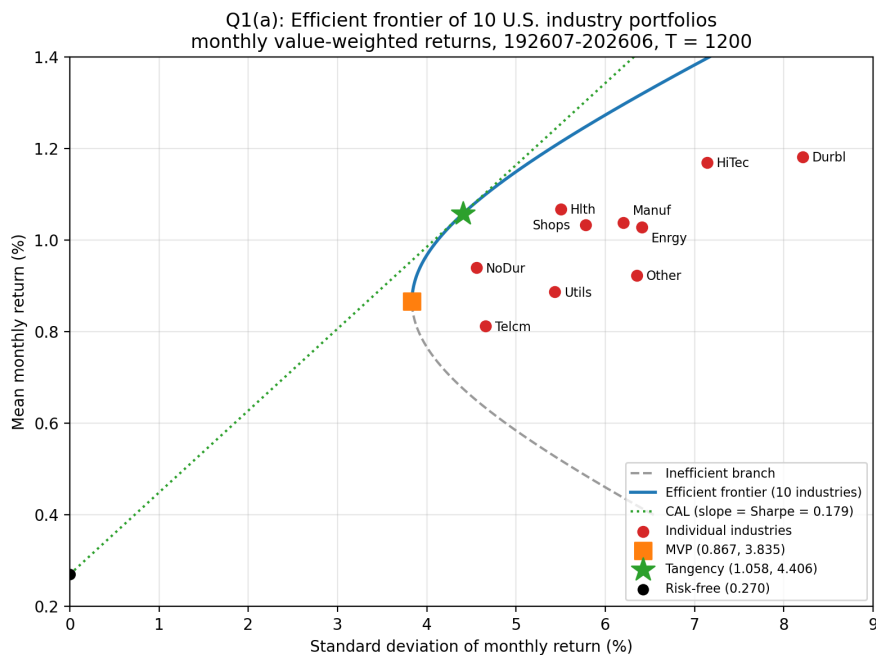


Figure 1: Efficient frontier of the ten industries in mean–standard deviation space. Red dots are the individual industries, the orange square is the minimum variance portfolio, the green star is the tangency portfolio, and the dotted line is the capital allocation line from the risk-free rate. The dashed grey curve is the inefficient lower branch.

Discussion of the weights.

- **NoDur is the largest long position in both portfolios**, 0.72 in the MVP and 0.66 in the tangency portfolio. It has the lowest standard deviation of the ten industries, 4.56% per month, and the lowest residual volatility once the other nine are controlled for. It earns that weight by hedging the movements the industries share.
- **The two portfolios diverge on Telcm**, which the MVP holds at 0.49 and the tangency portfolio at only 0.15. Telcm has the second-lowest standard deviation but the lowest mean return in the sample, 9.75% per year. Minimizing variance alone makes it valuable; once expected return enters, its low volatility no longer pays for the weakest risk premium of the ten.
- **The tangency portfolio is the more aggressive of the two**, despite being long in 8 of 10 industries against 5 for the MVP. Its gross leverage is 2.83 against 2.21 and its cross-sectional weight dispersion 0.359 against 0.309. The difference is where the extremity sits: the MVP concentrates on the long side and spreads small shorts across five industries, three of them negligible, while the tangency portfolio funds eight longs out of a single short of -0.69 in Other, the lowest-Sharpe industry. The extra leverage is what it costs to bet on mean differences that are small next to an average pairwise correlation of 0.696.

Shape of the frontier. Minimizing $w'\Sigma w$ subject to $w'\mu = m$ and $w'\mathbf{1} = 1$ gives first-order conditions that are linear in w , so the optimal weights are an affine function of the target mean,

$$w(m) = \frac{c - bm}{d} \Sigma^{-1} \mathbf{1} + \frac{am - b}{d} \Sigma^{-1} \mu, \quad (2)$$

with $a = \mathbf{1}'\Sigma^{-1}\mathbf{1}$, $b = \mathbf{1}'\Sigma^{-1}\mu$, $c = \mu'\Sigma^{-1}\mu$ and $d = ac - b^2$. Substituting back into the objective gives

$$\sigma^2(m) = \frac{am^2 - 2bm + c}{d}, \quad (3)$$

a quadratic in m , and therefore a parabola in mean–variance space and a hyperbola in the mean–standard deviation space of Figure 1. Here $a = 0.0680$ and $d = 0.000528$, both positive, so the parabola opens toward higher risk. Three things follow.

- **The curvature is diversification.** If the industries were perfectly correlated, combining them would reduce no risk and the attainable set would be a straight line. Correlations below one bend it leftward, and how far it bends measures how much diversification is available. Here the average pairwise correlation is 0.696, so the bend is modest: the MVP standard deviation of 3.83 is only 15.9% below NoDur’s 4.56, whereas part (c) shows it would fall to 1.82 if the industries were uncorrelated.
- **The vertex is the MVP.** Minimizing the quadratic gives $m = b/a = 0.8672$ and $\sigma^2 = 1/a = 14.70$, matching the minimum variance portfolio in Table 1 exactly.
- **Dominance makes only the upper branch efficient.** Because $\sigma^2(m)$ is quadratic, every variance above $1/a$ is attained by two portfolios, one on each side of the vertex. They carry identical risk but the lower one returns less, so it is dominated. Since investors prefer a higher expected return at the same risk, no one would ever hold it: the lower branch is feasible but never optimal, which is why it is dashed in Figure 1.

1.2 Question 1(b): Reliability of the Mean Return Estimates

The standard error of each sample mean is $\hat{\sigma}_i/\sqrt{T}$ with $T = 1200$.

Table 2: Reliability of the estimated mean returns, annualized. The t -statistic tests the null that the true mean return is zero.

Industry	Mean (%/yr)	Std. Error (%/yr)	t -statistic
NoDur	11.28	1.58	7.14
Durbl	14.19	2.85	4.99
Manuf	12.46	2.15	5.80
Enrgy	12.35	2.22	5.56
HiTec	14.04	2.47	5.67
Telcm	9.75	1.61	6.04
Shops	12.41	2.00	6.20
Hlth	12.82	1.91	6.73
Utils	10.65	1.88	5.66
Other	11.08	2.20	5.03

We then raise every estimated mean by one of its own standard errors, $\tilde{\mu}_i = \hat{\mu}_i + \hat{\sigma}_i/\sqrt{T}$, and recompute the tangency portfolio. The MVP is unchanged by construction because it does not use μ .

Table 3: Tangency portfolio weights and statistics before and after raising every estimated mean by one standard error.

Industry	Baseline	Means +1 SE	Change
NoDur	0.6645	0.5686	-0.0960
Durbl	0.1340	0.1526	0.0187
Manuf	-0.2242	-0.2611	-0.0370
Enrgy	0.2808	0.2864	0.0056
HiTec	0.1865	0.1781	-0.0084
Telcm	0.1525	0.1709	0.0184
Shops	0.0725	0.0671	-0.0054
Hlth	0.3530	0.3400	-0.0129
Utils	0.0721	0.0965	0.0244
Other	-0.6916	-0.5991	0.0926
<i>Portfolio mean</i> (%/mo)	1.0584	1.1980	0.1395
<i>Portfolio std. dev.</i> (%/mo)	4.4063	4.3771	-0.0292
<i>Sharpe ratio</i> (monthly)	0.1789	0.2120	0.0331

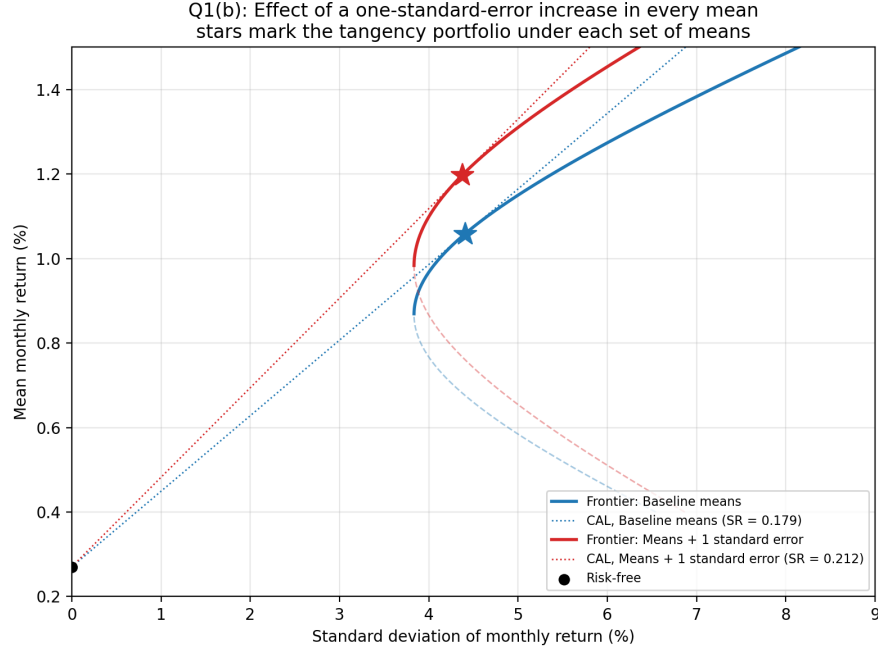


Figure 2: Efficient frontiers and capital allocation lines under the baseline means (blue) and after a one-standard-error increase in every mean (red). Stars mark the tangency portfolio in each case.

Discussion.

- **The means are not estimated reliably enough to rank the industries.** Every industry mean is significantly different from zero, with t -statistics from 4.99 to 7.14. Mean–variance optimization needs the ranking of the industries rather than the level, so we test all 45 pairwise differences instead, using the standard error of each paired difference series. Only one is significant at the 5% level (HiTec minus Telcm, $t = 2.31$), and 45 tests at that level would throw up about two false positives by chance. The cross-section therefore looks the same as ten industries with identical means. We then split the century at 1976 and compare the two halves. The rank correlation is 0.745, so the extremes do hold up, with Durbl and HiTec leading in both halves and Telcm trailing in both, while the middle reshuffles: NoDur moves from ninth to fifth and Manuf from fourth to seventh.
- **The tangency weights move only moderately.** Raising every mean by one standard error shifts the weight vector by 0.32 in total absolute terms, the largest single change is 0.096 in NoDur, and the cosine similarity between the two weight vectors is 0.9953. The composition tilts, and the direction of the tilt stays the same.
- **The frontier moves much more than the weights do.** The tangency Sharpe ratio rises from 0.1789 to 0.2120, an increase of 18.5%, and the capital allocation line shifts with it (Figure 2). So a perturbation that no test could detect relocates the whole opportunity set. We raised every mean at once, so a good part of this is the frontier shifting up as a whole, and only a smaller part comes from industries changing places. That is still the point: a century of data does not pin down even the level of the achievable Sharpe ratio.

1.3 Question 1(c): Reliability of the Covariance Matrix

We recompute the frontier with the covariances set to zero (keeping the estimated variances on the diagonal) and then with Σ replaced by the identity matrix. Table 4 reports each portfolio twice: once with risk measured by the covariance matrix used to build it, and once with the same weights evaluated under the true sample Σ . The second measurement is needed because the assumed risk units are not comparable across the three cases, and under the identity matrix they are arbitrary.

Table 4: Portfolio statistics under the three covariance assumptions. The mean vector μ is the same in every case. “Assumed” measures risk with the matrix used to construct the weights; “true” measures the same weights under the full sample covariance matrix. Means and standard deviations are monthly percentages.

Case	Mean	Assumed Σ		True Σ	
		Std. Dev.	Sharpe	Std. Dev.	Sharpe
Full sample — MVP	0.8672	3.8346	0.1557	3.8346	0.1557
Full sample — Tangency	1.0584	4.4063	0.1789	4.4063	0.1789
Diagonal — MVP	0.9794	1.8242	0.3888	4.8467	0.1463
Diagonal — Tangency	0.9960	1.8455	0.3933	4.9516	0.1466
Identity — MVP	1.0086	0.3162	2.3351	5.1422	0.1436
Identity — Tangency	1.0257	0.3199	2.3621	5.2675	0.1434

Table 5: Decomposition of portfolio variance into the contribution of the ten variance terms and the forty-five covariance terms, using the full sample covariance matrix. Monthly, in squared percent.

Portfolio	Total variance	From variances	From covariances	Covariance share
Equal weighted	26.442	3.741	22.701	85.9%
MVP	14.704	23.725	−9.021	−61.3%
Tangency	19.415	41.264	−21.849	−112.5%

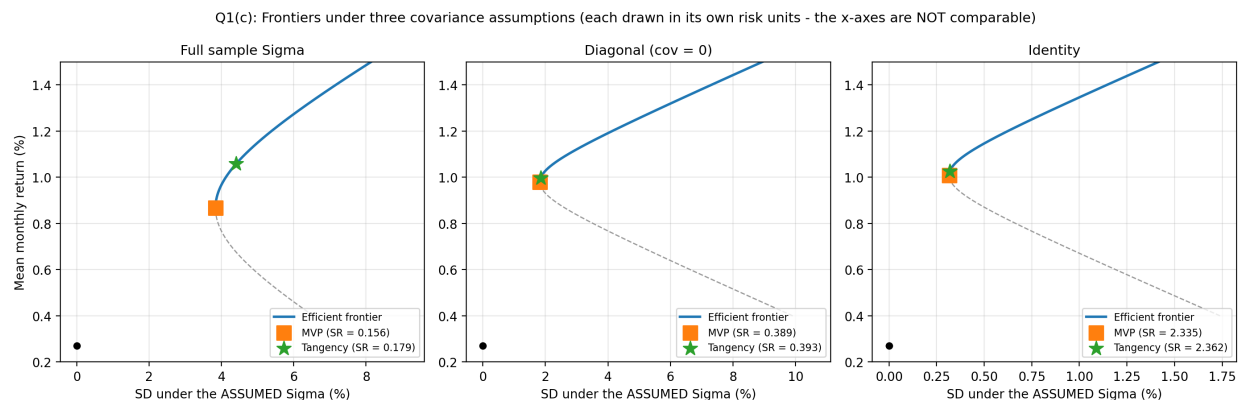


Figure 3: Efficient frontiers under the full sample, diagonal and identity covariance matrices, each drawn in its own risk units. The horizontal axes are therefore not comparable across panels.

Discussion.

- **The covariance matrix is estimated precisely, though it is not stable.** A mean estimate carries 17.2% relative error on average; a volatility carries 2.0%. Monthly returns bounce around five to seven times more than they drift upward, which makes the mean hard to pin down and leaves the volatility unaffected. Daily data would sharpen Σ further and would not help μ at all. Stability is a separate matter: splitting the century at 1976, average pairwise correlation falls from 0.779 to 0.581, so Σ is measured precisely and still moves. The two experiments below set 45 covariances to zero, which is far bigger than any estimation error, so they test what happens when we assume something wrong rather than when we measure something badly.
- **The frontier changes far more than in part (b).** Measured on its own assumed risk, the tangency Sharpe ratio goes from 0.1789 to 0.3933 under the diagonal matrix and to 2.3621 under the identity, against a move of 18.5% in part (b). We then take the same weights and measure risk with the real Σ , which gives 0.1789, 0.1466 and 0.1434. Dropping the covariances makes the portfolio look much better and leaves it slightly worse.
- **The covariance terms matter far more than the variance terms.** For an equal-weighted portfolio, 85.9% of total variance comes from the 45 covariance terms rather than the 10 variances. Two results in this section confirm that.

The first is the shape of the two misspecified frontiers, which look almost identical to each other. Reaching 0.2 above the MVP costs 2.10 times the MVP's own risk under the diagonal matrix and 2.04 under the identity, against 1.16 under the real Σ . Removing the correlations changes everything, and flattening the variances afterwards changes almost nothing. With real correlations the ten industries act like 2.26 independent bets, and with correlations removed they act like 10, which is why the real MVP variance of 14.70 is 4.4 times what it would be with no correlation.

The second is that the damage falls unevenly. Misspecifying Σ costs the MVP 6.0% and 7.8% of its true Sharpe ratio, and costs the tangency portfolio 18.1% and 19.8%. Under the real Σ the tangency portfolio beats the MVP by 14.9%, and under the diagonal and identity matrices that edge falls to 0.2% and -0.1% . Minimizing risk works roughly well with variances alone, so the MVP survives, while the tangency portfolio's whole edge comes from the covariances.

1.4 Question 1(d): Jorion Simulation under the Normal Distribution

Each simulation draws $T = 1200$ monthly return vectors from a multivariate normal distribution with the historical μ and Σ . The MVP and tangency weights are computed from the *simulated* sample and then applied to the *actual* industry returns, so the recorded mean and standard deviation are $w'\mu$ and $\sqrt{w'\Sigma w}$ at the historical moments. This is the essential feature of the Jorion (1986) design: the weights carry estimation error but are judged against the true distribution. The experiment is repeated 1,000 times.

Table 6: Estimation error across 1,000 simulations, for both Question 1(d) and Question 1(e). Portfolio means and standard deviations are evaluated on the actual industry returns, in percent per month. “SD of weights” is the cross-simulation standard deviation of a portfolio weight, averaged over the ten industries.

Experiment	Portfolio mean		Portfolio std. dev.		SD of weights
	Average	Std. dev.	Average	Std. dev.	
Normal — MVP	0.8671	0.0096	3.8489	0.0066	0.0381
Normal — Tangency	1.0650	0.0793	4.9938	0.6388	0.2716
Bootstrap — MVP	0.8693	0.0176	3.8627	0.0142	0.0523
Bootstrap — Tangency	1.0692	0.0802	5.0046	0.6377	0.2703

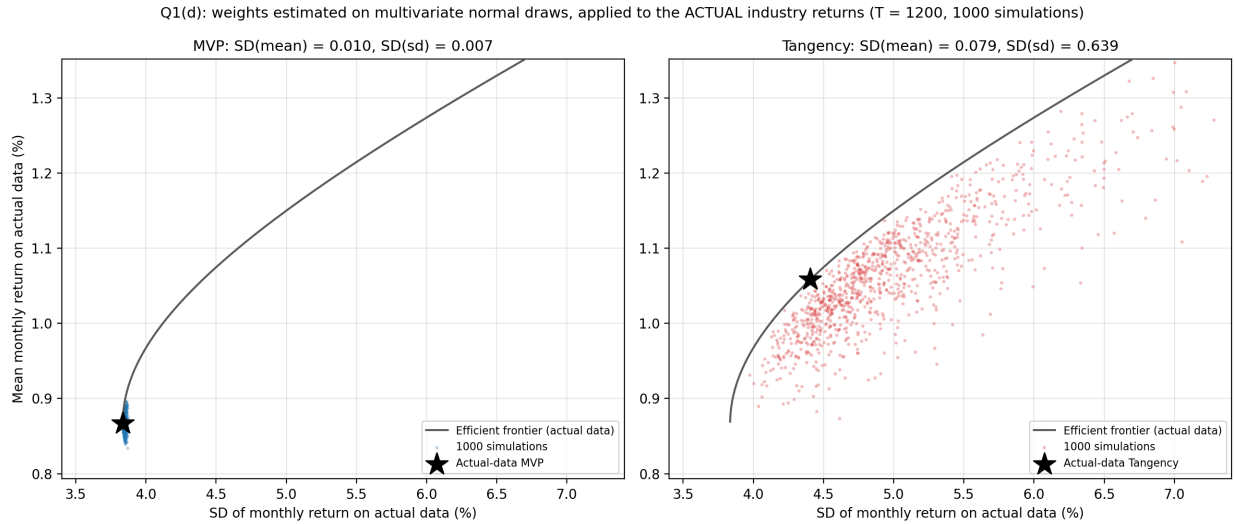


Figure 4: Question 1(d). Each point is one of 1,000 simulations: weights estimated from a multivariate normal sample of $T = 1200$ months, then applied to the actual industry returns. The black star is the portfolio built from the actual data. Both panels share the same axes.

Discussion. [TO WRITE: state which portfolio is estimated with less error, and why]

AI Integration — Simulation code generation. Prompt: [TO WRITE: paste the exact prompt given to the LLM]

What we had to fix or adjust: [TO WRITE: list the corrections]

Did the LLM apply the simulated weights to the actual returns? [TO WRITE: state explicitly whether the LLM made the in-sample mistake]

Did the LLM correctly predict which portfolio has smaller estimation error, and does its economic explanation hold up? [TO WRITE: evaluate the LLM’s prediction and reasoning]

1.5 Question 1(e): Empirical Row Bootstrap Simulation

We repeat the experiment under the empirical distribution. Each simulation draws $T = 1200$ month indices uniformly with replacement from $\{1, \dots, T\}$ and takes the corresponding rows of

the actual return matrix, so all ten industry returns for a sampled month stay together. Sampling whole rows preserves the contemporaneous cross-sectional dependence and the non-normal shape of the marginal distributions, while destroying serial dependence such as volatility clustering. As before, weights are estimated on the bootstrap sample and applied to the actual returns, 1,000 times. Results are in Table 6 alongside the normal simulation.

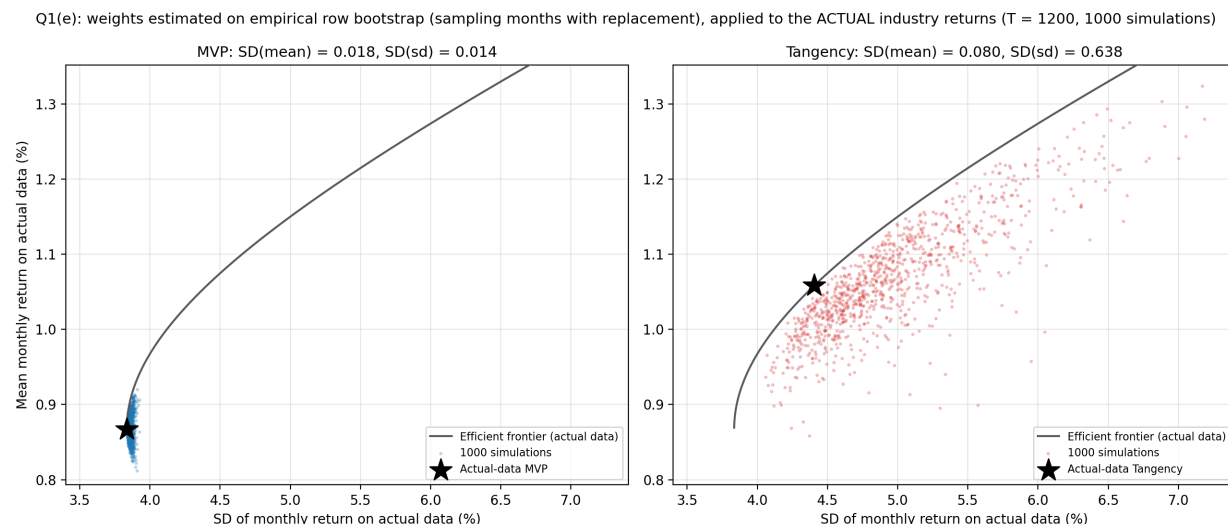


Figure 5: Question 1(e). Weights estimated on empirical row-bootstrap samples and applied to the actual industry returns, 1,000 simulations. Axes are identical to Figure 4 so the two experiments can be compared directly.

Discussion. [TO WRITE: compare estimation error under the bootstrap with the normal simulation of part (d)]